

Eloy Alvarado Narváez

Departamento de Estadística
Pontificia Universidad Católica de Chile
Santiago, Chile • Office 137



Employment

2026 –	Assistant Professor, Pontificia Universidad Católica de Chile	Santiago, Chile.
2023 – 2025	Academic Instructor, Universidad Técnica Federico Santa María	Santiago, Chile.
2022 – 2022	Lecturer, Universidad Técnica Federico Santa María	Santiago, Chile.
2021 – 2022	External Adviser, Instituto de Fomento Pesquero	Valparaíso, Chile.
2019 – 2022	Lecturer, Universidad de Valparaíso	Valparaíso, Chile.
2017 – 2017	Lecturer, Pontificia Universidad Católica de Valparaíso	Valparaíso, Chile.
2016 – 2017	Statistical Analyst, Equifax Inc.	Santiago, Chile. / Atlanta, US.

Education

2018 – 2022	Ph.D. in Statistics, Instituto de Estadística, Universidad de Valparaíso Advisors: Moreno Bevilacqua and Christian Caamaño	Valparaíso, Chile.
2013 – 2015	B.Sc in Statistical Science, Pontificia Universidad Católica de Valparaíso	Valparaíso, Chile.
2014	Exchange program in Korea University	Seoul, South Korea

Research Articles

1. E. Alvarado, C. Caamaño-Carrillo, and D. R. Godoy. *Weibull random fields for spatially dependent positive data: an application to mining haul roads*. In: *International Statistical Review* (2026). forthcoming.
2. H. Araya, F. Plaza-Vega, and E. Alvarado. *Drift parameter estimation for the noisy Ornstein-Uhlenbeck process driven by a Rosenblatt process*. In: *Statistics & Probability Letters* 237 (2026), p. 110803. DOI: 10.1016/j.spl.2026.110803.
3. H. Araya, F. Plaza-Vega, and E. Alvarado. *Drift parameter identification for the Ornstein-Uhlenbeck process driven by Ornstein-Uhlenbeck with small General Gaussian noise*. In: *Statistical Papers* 67.49 (2026). DOI: 10.1007/s00362-026-01804-3.
4. E. Alvarado, F. Plaza-Vega, C. Montenegro, and O. Saavedra. *Deep learning-based classification of species in central-southern fisheries in Chile*. In: *Latin American Journal of Aquatic Research* 53.3 (2025), pp. 411–424. ISSN: 0718-560X. DOI: 10.3856/vol53-issue3-fulltext-3318.
5. M. Bevilacqua, E. Alvarado, and C. Caamaño-Carrillo. *A flexible Clayton-like spatial copula with application to bounded support data*. In: *Journal of Multivariate Analysis* 201 (2024). Copula Modeling from Abe Sklar to the present day, p. 105277. ISSN: 0047-259X. DOI: <https://doi.org/10.1016/j.jmva.2023.105277>.
6. V. Beltrán, M. Flores, C. Sanzana, F. Muñoz-Sepúlveda, E. Alvarado, B. Venegas, J. C. Molina, S. Rueda-Velásquez, and A. von Marttens. *Tooth Loss and Caries Experience of Elderly Chileans in the Context of the COVID-19 Pandemic in Five Regions of Chile*. en. In: *International Journal of Environmental Research and Public Health* 20.4 (Jan. 2023), p. 3001. ISSN: 1660-4601. DOI: 10.3390/ijerph20043001.

7. V. Beltrán, P. A. Mardones, J. Díaz, E. Alvarado, and A. von Marttens. *TEGO: A new concept of teledentistry for the elderly through a web platform and mobile app in the context of the covid-19 pandemic*. In: *Journal of Oral Research* 1 (2022), pp. 1–8. DOI: 10.17126/joralres.2022.023.
8. V. Beltrán, A. von Marttens, P. Acuña-Mardones, C. Sanzana-Luengo, S. J. Rueda-Velásquez, E. Alvarado, M. Flores, A. Cerda, and B. Venegas. *Implementation of a Teledentistry Platform for Dental Emergencies for the Elderly in the Context of the COVID-19 Pandemic in Chile*. In: *BioMed Research International* 2022 (2022). DOI: 10.1155/2022/6889285.
9. C. Sanzana-Luengo, L. Díaz, L. Abarza, E. Alvarado, and V. Beltrán. *Teledentistry protocol for the elderly in the context of the COVID-19 pandemic through a web platform/mobile app: approach from the general dentist*. In: *Journal of Oral Research* (2022), pp. 1–8. DOI: 10.17126/joralres.2022.026.
10. M. Stehlík, J. Kiselák, A. Dinamarca, E. Alvarado, F. Plaza, F. Medina, S. Stehlíková, J. Marek, B. Venegas, A. Gajdoš, et al. *REDACS: Regional emergency-driven adaptive cluster sampling for effective COVID-19 management*. In: *Stochastic Analysis and Applications* (2022), pp. 1–35. DOI: 10.1080/07362994.2022.2033126.

Grants and Projects

- 2025 – 2027 Spatial extension of the Gumbel copula: applications to some non-Gaussian random fields, Fondecyt de Iniciación 2025 ANID No. 11250479.
- 2023 – 2024 Modeling and estimating massive point referenced spatial data, MATH-AMSUD AMSUD220041. Research Associate.
- 2018 – 2022 National Doctoral Degree, ANID No. 2018-21180953.
- 2020 – 2020 Semi-presential technological support platform for urgent and priority dental care for the elderly in the context of the COVID-19 pandemic in the Chilean population, ANID, Reference No. COVID0766, Chile. Collaborator.

Teaching

Pontificia Universidad Católica de Chile

EYP2307 Regression Analysis. Sem. I 2026

Universidad Técnica Federico Santa María

ICN292 Management Information Systems. Sem. I 2023; Sem. I–II 2024; Sem. I 2025

PII402 Stochastic Processes (Postgraduate). Sem. II 2023; Sem. I 2024; Sem. I 2025

IND163C Business Analytics. Sem. II 2022; Sem. I 2023

Universidad Andrés Bello

CIND211 Stochastic Networks and Systems. Sem. II 2025

ATDFI102 Introduction to programming. Sem. I 2024; Sem. I 2025

CIND217 Business intelligence. Sem. I 2024; Sem. I 2025

IICGP01 Business intelligence. Sem. II 2023

Universidad de Valparaíso

IEST422 Statistical Quality Control. Sem. II 2022

LFIS325 Statistics for Physical Sciences. Sem. II 2022; Sem. II 2021; Sem. II 2020

IECD411 Machine Learning. Sem. I 2022

IEST414 Time Series. Sem. I 2022

IECD415 Multivariate Methods. Sem. I 2022

IECD325 Linear Models and Experimental Design. Sem. II 2021

IENT412	Sampling Theory II.	<i>Sem. I–II 2021; Sem. I 2020</i>
IENT322	Sampling Theory I.	<i>Sem. I 2021; Sem. II 2020; Sem. II 2019</i>
IEC311	Probability and Statistics.	<i>Sem. I 2021; Sem. I 2020</i>
IENT213	Statistical Methods.	<i>Sem. II 2019</i>
Pontificia Universidad Católica de Valparaíso		
EST205	Probability and Statistics.	<i>Sem. I–II 2017</i>

Academic Service

2026 – 2027	Director, Board of Directors, Chilean Statistical Society (SOCHE).
2024 – 2024	Scientific Committee Member, XLVII National Conference on Statistics, organised by the Chilean Statistical Society.
2024 – 2025	Internship Coordinator, Department of Industries, Campus Vitacura, Universidad Técnica Federico Santa María.
2024 – 2025	Member of the academic committee for the development of a new undergraduate program: Bachelor's Degree in Data Science, Universidad Técnica Federico Santa María.

Students

2025	Industrial Engineering, Universidad Técnica Federico Santa María. Advisor: José Donoso; Wilson Godoy; Isadora Araya; Gabriela Estrada; Fernanda Pincheira; Benjamín Mardones.
2024	Industrial Engineering, Universidad Técnica Federico Santa María. Advisor: Cristóbal Retamal; Nicolás Bravo.
2024	Luis Riquelme (Master in Mathematics, Universidad del Bío-Bío). Co-Advisor.
2023	Ricardo Menares (Statistics and Data Science Engineering, Universidad de Valparaíso). Co-advisor.
2022	Cristóbal Collao (Statistical Engineering, Universidad de Valparaíso). Co-advisor.

Conference Presentations

Jun 2026	<i>“Archimedean-like spatial copulas for non-gaussian random fields”</i> . Poster, 12th International Conference on Spatio-Temporal Modelling (METMA XII), Zaragoza. Organizer: Universidad de Zaragoza
Mar 2026	<i>“Spatial copula models for non-gaussian random fields”</i> . Contributed session (Spatial processes and applications), 17th Latin American Congress of Probability and Mathematical Statistics (CLAPEM 17), Montevideo. Organizer: Latin American Society of Probability and Mathematical Statistics (SLAPEM)
Nov 2025	<i>“Copula-Based Spatial Random Fields: Theory and Applications”</i> . Universidad del Bío-Bío. Organizer: Universidad del Bío-Bío
Sep 2025	<i>“Non-Standard Profit Efficiency in Banking: A Copula Approach”</i> . Campus Vitacura Colloquium. Organizer: Universidad Técnica Federico Santa María
Mar 2025	<i>“2nd Workshop: Reproducible Research with R”</i> . Open Science Webinar USM. Organizer: Universidad Técnica Federico Santa María
Nov 2024	<i>“Weibull random fields through Clayton spatial copula An application to mining haul roads”</i> . III Conference on Geostatistics and Spatio-Temporal Statistics: Theory and Applications. Organizer: Universidad Técnica Federico Santa María
Nov 2024	<i>“Weibull random fields through Clayton spatial copula An application to mining haul roads”</i> . III Statistical Engineering Conference. Organizer: Universidad de Santiago de Chile
Oct 2024	<i>“Archimedean-like spatial copulas”</i> . XLVII National Conference on Statistics. Organizer: Chilean Statistical Society

- Oct 2024 *“Analysis of the Intention to Use ICT in Industrial Engineering: Integrating UTAUT and Structural Equation Models”*. XXXVI Chilean Congress of Engineering Education. Organizer: Chilean Society of Engineering Education
- Aug 2024 *“Workshop: Reproducible Research with R”*. Open Science Webinar USM. Organizer: Universidad Técnica Federico Santa María
- Jul 2024 *“Experiences in Open Science”*. Open Science USM. Organizer: Universidad Técnica Federico Santa María
- Jul 2024 *“Archimedean-like spatial copulas and their applications”*. Colloquium on Statistics and Data Science. Organizer: Pontificia Universidad Católica de Chile
- Nov 2023 *“Tools and Strategies for Reproducible Research”*. 3rd Open Science Webinar USM. Organizer: Universidad Técnica Federico Santa María
- Oct 2023 *“AI-based Feedback for Engineering Students”*. XXXV Chilean Congress of Engineering Education. Organizer: Chilean Society of Engineering Education
- Dec 2022 *“A flexible Clayton-like spatial copula with application to bounded support data”*. III Statistics Workshop: Graduate Contributions and Scientific initiation. Organizer: Chilean Statistical Society
- Oct 2015 *“Stochastics differential equations in mathematical finance: The Black-Scholes Model”*. XXI Valparaíso Statistical Week. Pontificia Universidad Católica de Valparaíso